

CHE 201 Fall 2019 HW 5

Laura Brito Uriostegui

TOTAL POINTS

91 / 100

QUESTION 1

11 15 / 15

✓ - **0 pts** Correct

- **8 pts** The v_1 & v_2 are incorrect
- **7 pts** The initial and final mass were wrong

QUESTION 2

22 14 / 15

- **0 pts** Correct
- **4 pts** Missing comments if work is done by or on the water vapor
- **4 pts** Missing identification of work as an area on the diagram
- **2 pts** Sketch 1 is wrong. $P_2 > P_1$
- **1.5 pts** Sketch 2 is wrong. $P_3 > P_1$
- **1.5 pts** Sketch 3 is wrong. $V_4 > V_1$
- **2 pts** Sketch 4 is wrong. $P_5 < P_1$
- **1 pts** Work 1-2 is the whole area under the curve
- **1 pts** Work 1-5 is the whole area under the curve
- **1 pts** Process 1-2 should be a curved line
- **1 pts** Process 1-5 should be a curved line
- ✓ - **1 pts** Missing the process line on the diagram for process 1-3
- **1 pts** Missing the process line on the diagram for process 1-4
- **4 pts** Missing sketches for processes 1-3 and 1-4
- **1 pts** The arrow in process 1-5 should be pointing down to indicate the direction of the process
- **2 pts** The initial state is water vapor and not saturated liquid water
- **2 pts** Missing comments about work for processes 1-3 and 1-4

QUESTION 3

33 20 / 20

✓ - **0 pts** Correct

- **4 pts** Did not multiply specific volume by mass
- **3 pts** The equation used for work would result in a negative value. It should be $(P_1V_1 - P_2V_2)/(n-1)$ or $(P_2V_2 - P_1V_1)/(1-n)$
- **20 pts** Missing answer
- **7 pts** Should have used the polytropic process relationship $PV^n = \text{constant}$ to calculate the volume in state 2
- **7 pts** Wrong calculation of the integral for work. The correct answer should be $W = 31.4 \text{ Btu}$ (work done by the system).
- **20 pts** Wrong Solution. The correct answer should be $W = 31.4 \text{ Btu}$ (work done by the system).
- **3 pts** Wrong calculation/unit conversion. The correct answer should be $W = 31.4 \text{ Btu}$ (work done by the system).

QUESTION 4

44 15 / 15

✓ - **0 pts** Correct

- **1 pts** $v_a \approx v_f(400^\circ\text{F}) = 0.01864 \text{ ft}^3/\text{lb}$ OR $v_a = 0.018334 \text{ ft}^3/\text{lb}$ (from Table A-5E)
- **1 pts** $h_a = h_f(400^\circ\text{F}) + v_f(400^\circ\text{F})[P - P_{\text{sat}}(400^\circ\text{F})] = 384.6 \text{ Btu/lb}$ OR $h_a = 378.50 \text{ Btu/lb}$ (from Table A-5E)
- **1 pts** Using Table A-10E and Eq 3.11, $v_b \approx v_f(95^\circ\text{F}) = 0.01371 \text{ ft}^3/\text{lb}$
- **1 pts** Using Table A-10E and Eq 3.14, $h_b \approx h_f(95^\circ\text{F}) = 42.47 \text{ Btu/lb}$
- **1 pts** Using Table A-13 and Eq 3.11, $v_c \approx v_f(20^\circ\text{C}) = 1.6386 \times 10^{-3} \text{ m}^3/\text{kg}$
- **1 pts** Using Table A-13 and Eq 3.14, $h_c \approx h_f(20^\circ\text{C}) = 274.26 \text{ kJ/kg}$
- **1 pts** Using Table A-17 and Eq 3.11, $v_d \approx v_f(0^\circ\text{C}) = 1.89 \times 10^{-3} \text{ m}^3/\text{kg}$

- **1 pts** Using Table A-17 and Eq 3.14, $h_d \approx h_f (0^\circ\text{C}) =$

95.1 kJ/kg

- **15 pts** Missing answer

- **1 pts** Missing v.a)

- **4 pts** Missing answer for d)

- **2 pts** Missing location of points b and c

- **15 pts** Missing answer

- **3 pts** Wrong answer for a)

- **4 pts** Missing answer for c)

- **7 pts** Wrong or missing T-v diagram sketches with the location of the points

- **10 pts** Energy balance equation was incorrect

QUESTION 5

5 5 12 / 20

- **0 pts** Correct

- **2 pts** Wrong or missing p-v diagram with sketch of the cycle

✓ - **5 pts** Missing or wrong calculation of Q_{in}

✓ - **3 pts** Missing or wrong calculation of thermal efficiency of the cycle. Correct answer is 12%

- **5 pts** Missing or wrong calculation of W_{cycle} or Q_{out}

- **2.5 pts** Missing calculation of V_3 and V_1 OR V_3 and T_3

- **1 pts** W_{3-1} is negative

- **1.5 pts** Correct equation but wrong solution for thermal efficiency. Correct answer is 12%

- **20 pts** Missing answer

- **1 pts** $T_3 = 887\text{K}$

- **2.5 pts** Missing or wrong calculation of T_2 , T_3 and P_2

- **1.5 pts** Did not show conversion to english units

- **10 pts** Correct solution process, but wrong values.

The correct answer for part a is 12%.

QUESTION 6

6 6 15 / 15

✓ - **0 pts** Correct

- **15 pts** Prob 3.61

- **15 pts** Final T ?

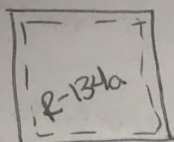
- **5 pts** T_f Calculation was incorrect

- **5 pts** Specific heat of copper incorrect

CHE 201 Hw 5

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Q3.16

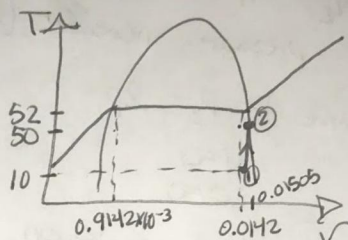


$T_1 = 10^\circ\text{C}$ liquid-vapor mixture
 $V_1 = 1.5\text{m}^3$

$T_2 = 50^\circ\text{C}$ $X = 100$ or 1.00
Phase $\Rightarrow$ liquid-vapor (Table A-10)

Find: $m_{\text{vapor}} (i) = ? \text{ kg}$
 $m_{\text{vapor}} (F) = ? \text{ kg}$

Model:
- Closed
- $V = \text{constant}$
- $\omega = 0$



$$\text{slope}_{T_2} = \frac{(0.0142 - 0.0159) \text{ m}^3/\text{kg}}{(52 - 48)^\circ\text{C}} = -4.25 \times 10^{-4} \text{ m}^3/\text{kg}\cdot^\circ\text{C} = \frac{(v_2 - 0.0159) \text{ m}^3/\text{kg}}{(50 - 48)^\circ\text{C}}$$

$$v_2 = 0.01505 \text{ m}^3/\text{kg}$$

$$m = \frac{V}{v_2} \rightarrow \frac{1.5 \text{ m}^3}{0.01505 \text{ m}^3/\text{kg}} = 99.67 \text{ kg}$$

$$X = \frac{m_{\text{vapor}_2}}{m} \rightarrow 1.00 = \frac{m_{\text{vapor}_2}}{99.67 \text{ kg}} \rightarrow \boxed{m_{\text{vapor}_2} = 99.67 \text{ kg}}$$

liquid:

$$\text{slope}_{T_1} = \frac{(0.7971 \times 10^{-3} - 0.7884 \times 10^{-3}) \text{ m}^3/\text{kg}}{(12 - 8)^\circ\text{C}} = 2.175 \times 10^{-6} \text{ m}^3/\text{kg}\cdot^\circ\text{C} = \frac{(v_1 - 0.7884 \times 10^{-3}) \text{ m}^3/\text{kg}}{(10 - 8)^\circ\text{C}}$$

$$v_1 = 7.9275 \times 10^{-4} \text{ m}^3/\text{kg}$$

gas:

$$\text{slope}_{T_1} = \frac{(0.0460 - 0.0525) \text{ m}^3/\text{kg}}{(12 - 8)^\circ\text{C}} = -0.001625 \text{ m}^3/\text{kg}\cdot^\circ\text{C} = \frac{(v_1 - 0.0525) \text{ m}^3/\text{kg}}{(10 - 8)^\circ\text{C}}$$

$$v_{g1} = 0.04925 \text{ m}^3/\text{kg}$$

$$v = v_f + X_1(v_g - v_f)$$

$$0.01505 \text{ m}^3/\text{kg} = (7.9275 \times 10^{-4}) + X_1(0.04925 - 7.9275 \times 10^{-4}) \text{ m}^3/\text{kg}$$

$$X_1 = 0.2942$$

$$\boxed{m_{\text{vapor}_1} = 29.3 \text{ kg}}$$

$$X = \frac{m_{\text{vapor}_1}}{m} \rightarrow 0.2942 = \frac{m_{\text{vapor}_1}}{99.67 \text{ kg}}$$

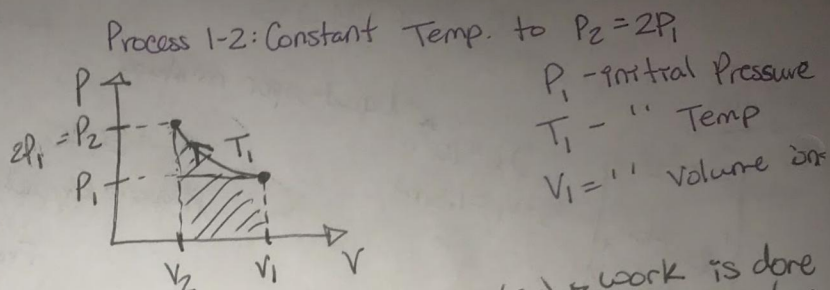
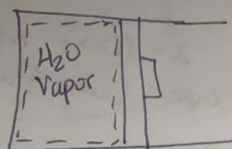
11 15 / 15

✓ - 0 pts Correct

- 8 pts The v_1 & v_2 are incorrect

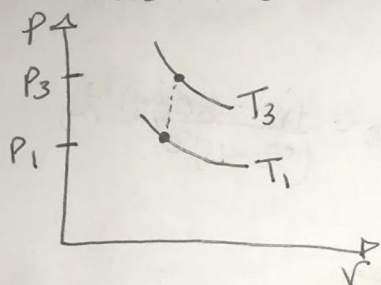
- 7 pts The initial and final mass were wrong

Q3.24



Volume is decreasing as the pressure increases (2x) → work is done on the system

Process 1-3: Constant volume to $P_3 = 2P_1$

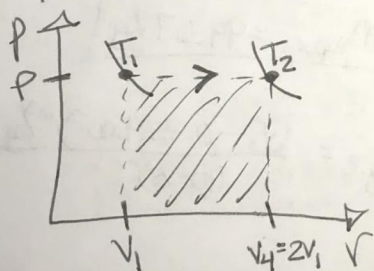


$$V=0$$

$$W=0$$

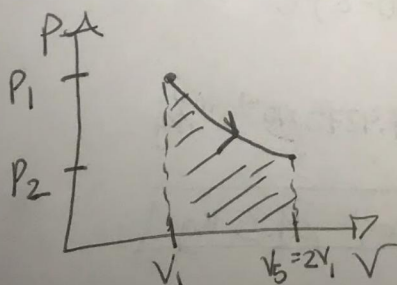
NO work on the system

Process 1-4: Constant pressure to $V_4 = 2V_1$



work is done by the system

Process 1-5: Constant Temperature to $V_5 = 2V_1$

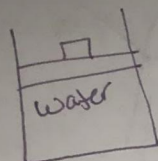


work is done by the system

22 14 / 15

- 0 pts Correct
- 4 pts Missing comments if work is done by or on the water vapor
- 4 pts Missing identification of work as an area on the diagram
- 2 pts Sketch 1 is wrong. $P_2 > P_1$
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- 1 pts Process 1-2 should be a curved line
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- ✓ - 1 pts Missing the process line on the diagram for process 1-3
- 1 pts Missing the process line on the diagram for process 1-4
- 4 pts Missing sketches for processes 1-3 and 1-4
- 1 pts The arrow in process 1-5 should be pointing down to indicate the direction of the process
- 2 pts The initial state is water vapor and not saturated liquid water
- 2 pts Missing comments about work for processes 1-3 and 1-4

Q3.26



$$\begin{aligned} m &= 0.5 \text{ lb} \\ P_1 &= 40 \text{ lbf/in}^2 \\ T_1 &= 300^\circ\text{F} \\ V_1 &= 11.04 \text{ ft}^3/\text{lb} \end{aligned}$$

$$P_2 = 14.7 \text{ lbf/in}^2$$

Model

- Polytropic process $pr^{1.2} = \text{const.}$
 $\hookrightarrow$ superheated water vapor

$$P = \frac{\text{constant}}{V^n}$$

$$P_2 V_2^{1.2} = P_1 V_1^{1.2} = \text{constant}$$

$$14.7 \text{ lbf/in}^2 V_2^{1.2} = 40 \text{ lbf/in}^2 (11.04 \text{ ft}^3/\text{lb})^{1.2}$$

$$V_2^{1.2} = 48.563 \rightarrow V_2 = 25.42 \text{ ft}^3/\text{lb}$$

$$\omega = \int_{V_1}^{V_2} P dV = \int_{V_1}^{V_2} \frac{\text{constant}}{V^n} dV \rightarrow \frac{(\text{constant}) V_2^{1-n} - (\text{constant}) V_1^{1-n}}{1-n} = \frac{P_2 V_2 - P_1 V_1}{1-n}$$

$$V = m \cdot v$$

$$\omega = \frac{0.5 \text{ lb} [(14.7 \text{ lbf/in}^2)(25.42 \text{ ft}^3/\text{lb}) - (40 \text{ lbf/in}^2)(11.04 \text{ ft}^3/\text{lb})]}{1-1.2} \left(\frac{144 \text{ in}^2}{1 \text{ ft}^2} \right) \left(\frac{1 \text{ Btu}}{778.17 \text{ ft-lbf}} \right)$$

$$\omega = 24942.43 \text{ lbf-ft}$$

$$\boxed{\omega = 32.05 \text{ Btu}}$$

33 20 / 20

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- 3 pts The equation used for work would result in a negative value. It should be $(P_1V_1 - P_2V_2)/(n-1)$ or $(P_2V_2 - P_1V_1)/(1-n)$

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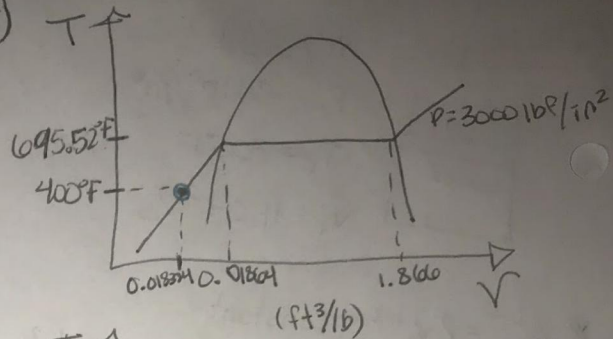
Q3.31 a) Water (Table A-3E)

Given: $T = 400^\circ\text{F}$
 $P = 3000 \text{ lbf/in}^2$

Find & sketch:

$$v = 0.018334 \text{ ft}^3/\text{lb}$$

$$h = 378.50 \text{ Btu/lb}$$



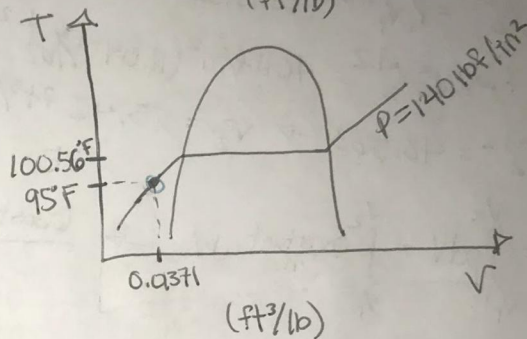
b) Refrigerant 134a

Given: $T = 95^\circ\text{F}$
 $P = 150 \text{ lbf/in}^2$

Find & sketch:

$$v = 0.01371 \text{ ft}^3/\text{lb}$$

$$h = 42.47 \text{ Btu/lb}$$



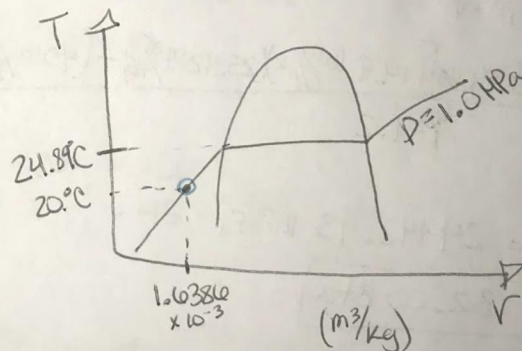
c) Ammonia

Given: $T = 20^\circ\text{C}$
 $P = 1.0 \text{ MPa}$

Find & sketch:

$$v = 1.6386 \times 10^{-3} \text{ m}^3/\text{kg}$$

$$h = 274.26 \text{ kJ/kg}$$



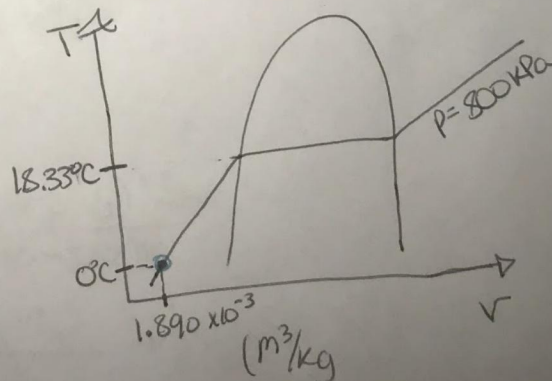
d) Propane

Given: $P = 800 \text{ kPa}$
 $T = 0^\circ\text{C}$

Find & sketch:

$$v = 1.890 \times 10^{-3} \text{ m}^3/\text{kg}$$

$$h = 95.1 \text{ kJ/kg}$$

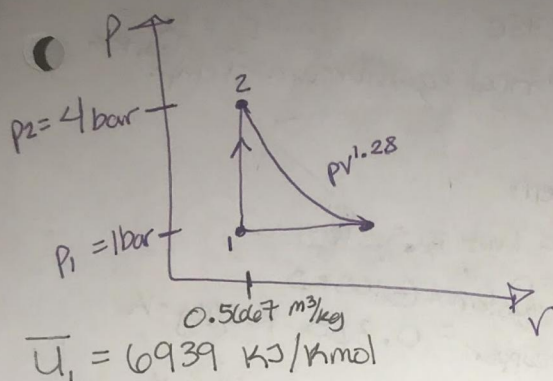


4 4 15 / 15

✓ - 0 pts Correct

- 1 pts $v_a \approx v_f (400^\circ\text{F}) = 0.01864 \text{ ft}^3/\text{lb}$ OR $v_a = 0.018334 \text{ ft}^3/\text{lb}$ (from Table A-5E)
- 1 pts $h_a = h_f (400^\circ\text{F}) + v_f (400^\circ\text{F}) [P - P_{\text{sat}} (400^\circ\text{F})] = 384.6 \text{ Btu/lb}$ OR $h_a = 378.50 \text{ Btu/lb}$ (from Table A-5E)
- 1 pts Using Table A-10E and Eq 3.11, $v_b \approx v_f (95^\circ\text{F}) = 0.01371 \text{ ft}^3/\text{lb}$
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- 1 pts Using Table A-17 and Eq 3.14, $h_d \approx h_f (0^\circ\text{C}) = 95.1 \text{ kJ/kg}$
- 15 pts Missing answer
- 1 pts Missing v.a)
- 4 pts Missing answer for d)
- 2 pts Missing location of points b and c
- 15 pts Missing answer
- 3 pts Wrong answer for a)
- 4 pts Missing answer for c)
- 7 pts Wrong or missing T-v diagram sketches with the location of the points

Q 3.97



$$m = 2 \text{ kg CO}_2$$

$$P_1 = 1 \text{ bar}$$

$$T_1 = 300 \text{ K}$$

Model

- Power cycle
- $\Delta KE = \Delta PE = 0$
- CO₂ properties
- Assuming ideal gas

State 1

$$P_1 V_1 = m R_{\text{CO}_2} T_1$$

$$\frac{V_1}{m} = \frac{R_{\text{CO}_2} T_1}{P_1} \rightarrow V_1 = \frac{(0.1889 \frac{\text{kJ}}{\text{kg} \cdot \text{K}}) \left(\frac{1000 \text{ N} \cdot \text{m}}{1 \text{ kJ}} \right) (300 \text{ K})}{(1 \text{ bar}) \left(\frac{10^5 \text{ N/m}^2}{1 \text{ bar}} \right)}$$

$$V_1 = 0.5667 \text{ m}^3/\text{kg}$$

$$V_1 = V_2$$

State 2 $\rightarrow V_2 = 0.5667 \text{ m}^3/\text{kg}$

Constant volume:

$$\frac{P_1}{T_1} = \frac{P_2}{T_2} \rightarrow \frac{1 \text{ bar}}{300 \text{ K}} = \frac{4 \text{ bar}}{T_2}$$

$$T_2 = 1200 \text{ K}$$

$$u_2 = 43,871 \text{ kJ/kmol}$$

State 3:

$$P_3 = P_1 \rightarrow P_3 = 1 \text{ bar}$$

$$P_2 V_2^{1.28} = P_3 V_3^{1.28} \rightarrow (4 \text{ bar}) (0.5667^{1.28}) = (1 \text{ bar}) (V_3^{1.28}) \Rightarrow V_3 = 1.6738 \text{ m}^3/\text{kg}$$

Calculate Temp. 3

$$\frac{T_3}{T_2} = \left(\frac{P_3}{P_2} \right)^{\frac{n-1}{n}} \rightarrow \frac{T_3}{1200 \text{ K}} = \left(\frac{1}{4} \right)^{\frac{1.28-1}{1.28}} \rightarrow T_3 = 886.1 \text{ K}$$

Specific internal energy:

$$\frac{(29476 - 29031) \frac{\text{kJ}}{\text{kmol}}}{(890 - 886) \text{ K}} = \frac{(29031 - u_3) \frac{\text{kJ}}{\text{kmol}}}{(890 - 886) \text{ K}}$$

$$u_3 = 29,298 \text{ kJ/kmol}$$

$$w_{1-2} = 0 \text{ kJ}$$

$$w_{2-3} = \frac{m R_{\text{CO}_2} (T_3 - T_2)}{1-n} \rightarrow \frac{(2 \text{ kg}) (0.1889 \times 10^3) (886.1 - 1200)}{1 - 1.28}$$

$$w_{2-3} = 423.68 \text{ kJ}$$

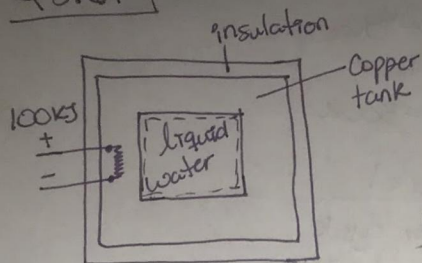
TABLE A-22 Ideal Gas Properties of Air (Continued)

$$\begin{aligned} w_{3-1} &= mP(v_1 - v_2) \\ &= mR(T_1 - T_3) \end{aligned} \rightarrow (2 \text{ kg})(0.1889)(300 - 886) \rightarrow w_{3-1} = -221.4 \text{ kJ}$$

5 5 12 / 20

- 0 pts Correct
- 2 pts Wrong or missing p-v diagram with sketch of the cycle
- ✓ - 5 pts Missing or wrong calculation of Q_{in}
- ✓ - 3 pts Missing or wrong calculation of thermal efficiency of the cycle. Correct answer is 12%
- 5 pts Missing or wrong calculation of W_{cycle} or Q_{out}
- 2.5 pts Missing calculation of V_3 and V_1 OR V_3 and T_3
- 1 pts W_{3-1} is negative
- 1.5 pts Correct equation but wrong solution for thermal efficiency. Correct answer is 12%
- 20 pts Missing answer
- 1 pts $T_3 = 887K$
- 2.5 pts Missing or wrong calculation of T_2 , T_3 and P_2
- 1.5 pts Did not show conversion to english units
- 10 pts Correct solution process, but wrong values. The correct answer for part a is 12%.

Q3.61



Copper tank

$$m_c = 13 \text{ kg}$$

$$T_i = 27^\circ\text{C}$$

Find $\Rightarrow$ Final equilibrium Temp.

liquid water

$$m_w = 4 \text{ kg}$$

$$T_2 = 50^\circ\text{C}$$

$$323.15 \text{ K}$$

Model

- Closed system

- $Q=0 \rightarrow$ No heat in system

- $\Delta KE = \Delta PE = 0$

- work on the system = -100 kJ

$$C_{\text{copper}} = 0.385 \text{ kJ/kg}\cdot\text{K}$$

$$C_{\text{water}} = 4.179 \text{ kJ/kg}\cdot\text{K}$$

Energy balance:

$$\Delta E = Q - W$$

$$\Delta KE + \Delta PE + \Delta U = Q - W$$

$$\Delta U = -W$$

$$(\Delta U_{\text{water}} - \Delta U_{\text{tank}}) = -W$$

$$C = \frac{du}{dT} \Rightarrow \Delta U = C \Delta T = C \Delta T$$

$$\Delta U = m C \Delta T = -W$$

Energy balance:

$$m_c \times C_c (T_f - T_{c,i}) + m_w \times C_w (T_f - T_{w,i}) = -W$$

$$(13 \text{ kg})(0.385 \text{ kJ/kg}\cdot\text{K})(T_f - 300.15 \text{ K}) + (4 \text{ kg})(4.179 \text{ kJ/kg}\cdot\text{K})(T_f - 323.15 \text{ K}) = 100 \text{ kJ}$$

$$21.721 T_f = 1344.085 - 273.15 \text{ C}$$

$$21.721 T_f = 1070.935$$

$$T_f = 49.30^\circ\text{C}$$

6 6 15 / 15

✓ - 0 pts Correct

- 15 pts Prob 3.61

- 15 pts Final T?

- 5 pts Tf Calculation was incorrect

- 5 pts Specific heat of copper incorrect

- 10 pts Energy balance equation was incorrect